II श्री गणेश प्रसन्न II

CAMPBELL-WALSH-WEIN · BOOK 15

Renal, Nephrology & Transplant

Concise rapid revision · facts answered in Papers 1–4 point to that answer

highlighted number to memorise · KEY POINT and ALGORITHM boxes carry the reasoning · P2 = covered in full in that Paper's answer book; the list under it gives what that answer misses

Chapter	Topic
Ch 88	Surgical, Radiologic and Endoscopic Anatomy of the Kidney and Ureter
Ch 90	Fundamentals of Nephrology
Ch 91	Renal Transplantation
★	Putting it together: kidney function and transplant decisions

88 Surgical, Radiologic and Endoscopic Anatomy of the Kidney and Ureter

1 · Kidney: position and relations

The kidney's neighbours are what you injure at surgery.

- The kidney is ≈**11 cm** long, 125–170 g (male), at T12–L3. The **right lies 1–2 cm lower**; both descend ≈3 cm on inspiration or standing.

P1 Embryology and anatomy of kidney; nephron-sparing surgery in small renal mass and renal mass — relations, coverings, Gerota open below, hilum order

Not in the paper answer:

- Traction on the hepatorenal or splenorenal ligament tears the liver or splenic capsule.

P1 Pelvicalyceal (renal collecting system) anatomy, its relevance to PCNL and endourology, and mini-PCNL — access hazards: pleura, lung, liver/spleen, retrorenal colon

Not in the paper answer:

- Upper-pole excision through the diaphragm risks pneumothorax.

2 · Coverings and compartments

- Each perirenal space is separate, so extravasation rarely crosses the midline. The **anterior pararenal space** (colon, duodenum, pancreas) does cross it; the posterior pararenal space holds fat only.
- Transperitoneally, open the **white line of Toldt**. Reach the right hilum by a **Kocher manoeuvre**; on the left, reflect the pancreatic tail, spleen and splenic vessels medially.

3 · Parenchyma and its imaging

- Cortex is ≈1 cm. **Columns of Bertin** carry interlobar arteries, so the safe track is **through pyramid and papilla**.
- Glomerular filtration rate (GFR) is ≈**125 mL/min**; ≈99% of filtrate is reabsorbed.

Modality	Normal kidney
Ultrasound	Cortex iso-/hypoechoic to liver (hyperechoic in newborn)
Computed tomography (CT)	Unenhanced 30–60 Hounsfield units (HU), cortex = medulla; enhanced 80–120 HU
Magnetic resonance (MRI)	T1 cortex brighter; T2 cortex slightly darker

CT phase	Timing	Use
Arterial	20–30 s	Vessels
Corticomedullary	30–70 s	Cortex enhances
Nephrographic	**80–120 s**	**Best for renal masses**
Excretory	>3 min	Collecting system, ureter

- On intravenous urography, no excretion at 24 h means a non-functioning kidney.
- **Corticomedullary differentiation** is lost in chronic kidney disease and acute kidney injury and returns with recovery.

Congenital anomalies

- **Malrotation** shows a central pelvis with calyces medial to it. In **ectopia**, poor drainage causes stasis, infection and stones.
- **Horseshoe**: lower poles fuse and ascent is arrested by the inferior mesenteric artery (**IMA**); it brings ureteropelvic junction obstruction (UPJO), stones, infection and multiple arteries.

4 · Renal vessels, lymphatics and nerves

End arteries make partial nephrectomy possible.

P2 Renovascular hypertension and ischaemic nephropathy: renal vascular anatomy, pathophysiology, evaluation and treatment — segmental end arteries, accessory 25–28%, Brödel line, veins; left renal vein **6–10 cm**, nutcracker, circumaortic

Not in the paper answer:

- **CT angiography** is the gold standard for vascular anatomy but misses vessels <2 mm.
- Right-sided lymph drains to interaortocaval/paracaval nodes but **may cross to left para-aortic** nodes.

5 · Percutaneous puncture and vessels

P1 Pelvicalyceal (renal collecting system) anatomy, its relevance to PCNL and endourology, and mini-PCNL — Sampaio puncture table: upper infundibulum 67%, fornix no arterial injury

Not in the paper answer:

- A through-and-through puncture hits anterior segmental vessels and causes marked haemorrhage.

> **KEY POINT**
> Why the fornix? Arteries run on both faces of the infundibula, and a forniceal puncture through the papilla avoids them. If an infundibulum is unavoidable, choose the lower pole. Vascular injury causes haemorrhage, arteriovenous fistula, pseudoaneurysm and loss of function.

6 · Pelvicalyceal system

P1 Pelvicalyceal (renal collecting system) anatomy, its relevance to PCNL and endourology, and mini-PCNL — Sampaio types, duplex pelvis, perpendicular and crossed calyces

Not in the paper answer:

- Dilatation is mimicked by an extrarenal pelvis, **megacalycosis** and a calyceal diverticulum.

7 · Ureter: course and relations

Injuries occur at crossings; stones stick at narrowings.

- The adult ureter is 22–30 cm. Stones impact at three **narrowings**: ureteropelvic junction (UPJ), iliac crossing and ureterovesical junction (UVJ).
- It lies on psoas and crosses the **common iliac bifurcation** (landmark), with gonadal vessels in front. The **pelvic brim** is a commonest injury site.
- In females it passes under the **uterine artery 1.5–2 cm** from the cervix ("water under the bridge"). In males the vas loops over it.
- The **intramural** segment is 1.2–2.5 cm long. Bladder pressure closes this oblique tunnel, which prevents reflux.

8 · Radiologic anatomy of the ureter

- Because of **peristalsis**, the whole ureter is rarely opacified; a continuous column suggests obstruction or atony.
- A stricture needs **proximal dilatation**; crossing vessels and iliac indentation can mimic it.
- Incomplete duplication (one bud, one orifice) is commoner than complete duplication (two buds, two orifices).

KEY POINT

Weigert–Meyer rule: the upper-moiety orifice is **inferior and medial** and obstructs (ureterocele); the lower moiety refluxes (vesicoureteral reflux, VUR).

- Bilateral medial displacement suggests retroperitoneal fibrosis or pelvic lipomatosis.

9 · Ureteric blood supply, nerves, histology

Segment	Artery	Enters	Endoureterotomy
Proximal	Renal	Medially	**Lateral**
Middle	Common iliac	Posteriorly	**Anterior**
Distal	Superior vesical	Laterally	**Medial**

- The longitudinal anastomosis is fragile, so preserve adventitia and avoid lateral retraction to prevent ischaemic stricture.
- Peristalsis is **myogenic**, so denervation does not cause reflux. Pain (T11–L1) refers to flank, groin, scrotum or labia.
- In **UPJO**, the junction has excess longitudinal muscle and collagen.

10 · Endoscopic anatomy

- The trigone is the bladder's most vascular part. Orifices lie 1–2 cm from the midline and move laterally on filling.

Lyon axis	Grades
Position	A medial trigone (normal) · B · C lateral
Configuration	0 volcano (normal) · 1 stadium · 2 horseshoe · 3 golf-hole

- A higher grade means more reflux. The **intramural ureter is narrowest** (3–4 mm) and may need dilatation.
- The lower-pole infundibulum needs ≈ **140°** deflection.

ALGORITHM

1. Infundibulotomy: incise superior, then inferior, then posterior quadrants.
2. **Never anterior**.

90 Fundamentals of Nephrology

1 · Why a urologist needs nephrology

- Each kidney has ≈**1 million** **nephrons**, fixed at birth. When nephron mass falls (nephrectomy, agenesis), the rest **hyperfilter**.
- Interstitial fibroblasts make **erythropoietin**; the proximal tubule makes **calcitriol** by 1α-hydroxylation.

> **KEY POINT**
> The glomerulus is a pressure filter set by arteriolar tone; the tubule is energy-hungry and near-hypoxic. Most disorders fail one or the other.

2 · Measuring kidney function

Serum creatinine (SCr) often misleads in surgical patients.

P2 GFR estimation: clinical assessment and in the prospective transplant donor — inulin/iohexol gold standard; creatinine secreted (24-h clearance overestimates); CKD-EPI 2021 race-free; cystatin C; combined equation most accurate

Not in the paper answer:

- SCr is falsely low with small muscle mass, and falsely raised by **trimethoprim and cimetidine**, which block secretion. Cystatin C is raised by hyperthyroidism, steroids and inflammation.

> **KEY POINT**
> Trap: after cystectomy or bladder repair, a rising SCr with ascites may be an **intraperitoneal urine leak** reabsorbed across the peritoneum, not kidney injury.

P1 Proteinuria — normal <**150 mg/day**, nephrotic >3.5 g/day; spot early-morning ACR first, PCR for non-albumin protein; dipstick misses light chains

Not in the paper answer:

- **eGFR is invalid when function is changing**, so acute kidney injury (AKI) is defined by SCr change and urine output.

3 · Acute kidney injury

Glomerular haemodynamics

- Net filtration pressure ≈**10 mm Hg** = glomerular hydrostatic 60 − Bowman's 18 − oncotic 32.

Arteriolar change	Renal blood flow	GFR
Afferent constriction	Falls	Falls
Afferent dilatation	Rises	Rises
Efferent constriction	Falls	Rises
Efferent dilatation	Rises	Falls

Angiotensin II blockade therefore drops GFR when perfusion is low.

Definition and staging (KDIGO)

Stage	Serum creatinine	Urine output
1	Rise ≥ **0.3 mg/dL** (within 48 h†) or ×1.5 baseline (within 7 days†)	<0.5 mL/kg/h for 6 h
2	×2 baseline	<0.5 mL/kg/h for 12 h
3	×3, or SCr ≥4 mg/dL with acute rise ≥0.5 (†or dialysis started)	<0.3 mL/kg/h for 24 h, or anuria 12 h

P2 Acute renal failure (AKI) and acute tubular necrosis — causes, ATN mechanism, AEIOU dialysis indications

Not in the paper answer:

- **3 months** separates acute kidney disease (AKD) from CKD.

Prerenal AKI

- Urine is bland, fractional excretion of sodium (FeNa) is low, and function **recovers immediately** on correction.
- **Autoregulation**: the **myogenic reflex** dilates the afferent arteriole via prostaglandins, so it is **disabled by NSAIDs** (non-steroidal anti-inflammatory drugs).
- In **tubuloglomerular feedback (TGF)**, angiotensin II constricts the efferent arteriole, so it is **disabled by renin–angiotensin–aldosterone system (RAAS) blockade**.
- **Modifiable** factors are hypotension, anaemia, sepsis, nephrotoxins, rhabdomyolysis and hypercalcaemia.

P2 Acute renal failure (AKI) and acute tubular necrosis — prerenal vs ATN indices: urine Na 20, FeNa < **1%**, FeUrea < **35%**, SG, muddy-brown casts, fluid response

Not in the paper answer:

- High FeNa after diuretics does not prove ATN; use fractional excretion of urea (FeUrea). **Response to correction is the single most important test.**
- Keep mean arterial pressure (MAP) > **65 mm Hg**. **Fluids** (mOsm/L): plasma and 0.9% saline 308; Plasmalyte 294; Ringer's lactate 273; 0.45% saline 154; 5% dextrose (D5) 0, so D5 is free water.

Intrinsic AKI: the cast localises the compartment

	Acute tubular necrosis (ATN)	Acute glomerulonephritis (GN)	Acute interstitial nephritis (AIN)
Cause	Ischaemia, toxins	Immune complex, anti-glomerular basement membrane (anti-GBM), cell-mediated	Drug, infection, systemic
Urine	Granular casts	Dysmorphic red cells, **red cell casts**, proteinuria	White cells, **white cell casts**, with or without eosinophils

Acute tubular necrosis

- **Rhabdomyolysis** (obese patient, long flexed lateral robotic case) needs **early aggressive isotonic fluid**; alkalinisation is unproven. †Prevent it with padding and limited flexion time.
- **Give diuretics only for symptomatic overload; they do not treat ATN.**

Postcontrast AKI

- This is an SCr rise within **48 h** of intravenous (IV) iodinated contrast. Risks include CKD, diabetes and the arterial route.

- **IV fluid before contrast is the only effective prevention** (†for eGFR <30 or AKI); use the lowest dose.
- **Gadolinium** groups I and III cause nephrogenic systemic fibrosis (**NSF**), so avoid them in AKI, eGFR <**30** and dialysis. Group II (gadobenate, gadoteridol, gadobutrol, gadoterate) is allowed. **Group III (gadoxetate) is treated as group I.**

Acute interstitial nephritis

- **Drugs cause most**, notably β-lactams, NSAIDs, proton pump inhibitors and **checkpoint inhibitors**.
- Suspect it in unexplained hospital AKI with **sterile pyuria**. Urine eosinophils are insensitive and non-specific MCQ.
- **Stop the drug**; early steroids may help, and about half recover fully.

Glomerulonephritis

- In microhaematuria, a rising SCr, proteinuria, red cell casts, hypertension or purpura call for **urgent nephrology and biopsy**, not cystoscopy first.

Complications of AKI

- **Overload** is urgent only with respiratory distress. Give a high-dose IV loop diuretic and add metolazone or acetazolamide (**sequential blockade**).
- For uraemic bleeding, give desmopressin (**DDAVP**) **0.3 µg/kg** IV/IM 30–60 min before the procedure.

Renal replacement therapy

- AEIOU (acidosis, electrolytes, ingestions, overload, uraemia) means failed medical management, not a biochemical threshold. **Early dialysis gives no benefit.** Continuous therapy is not superior.
- Non-recovery for ≥**90 days** means CKD or end-stage kidney disease (ESKD).

4 · Drug dosing in reduced kidney function

- In AKI, eGFR overestimates function, so reassess doses daily.
- †**Cockcroft–Gault** = (140 − age) × weight (kg) / (72 × SCr mg/dL), × 0.85 if female.
- Avoid NSAIDs; †withhold RAAS blockers in dehydration or perioperative hypotension ("sick-day" rule).
- †Avoid nitrofurantoin if eGFR <30 (UK guidance allows short courses at 30–44). †Stop metformin at eGFR <30, and withhold it before iodinated contrast in AKI or eGFR <30.

5 · Chronic kidney disease

- **KDIGO** (Kidney Disease: Improving Global Outcomes) defines CKD as abnormal structure or function for ≥**3 months**. Classify by **cause, GFR and albuminuria**.

GFR (mL/min/1.73 m²)	A1 albumin-to-creatinine ratio (ACR) <30 mg/g (<3 mg/mmol)	A2 30–300	A3 >300 (>30 mg/mmol)
G1 ≥90	Low*	Moderate	High
G2 60–89	Low*	Moderate	High
G3a 45–59	Moderate	High	Very high
G3b 30–44	High	Very high	Very high

GFR (mL/min/1.73 m²)	A1 albumin-to-creatinine ratio (ACR) <30 mg/g (<3 mg/mmol)	A2 30–300	A3 >300 (>30 mg/mmol)
G4 15–29	Very high	Very high	Very high
G5 <15 (kidney failure)	Very high	Very high	Very high

*G1–G2 count as CKD only with albuminuria or structural damage. Albuminuria can shift risk two bands: eGFR 43 with ACR 1100 mg/g is **G3bA3, very high**.

- **Each perioperative AKI is a modifiable progression factor**.

P2 Metabolic abnormalities in CKD — complications table: anaemia (Hb **10–11.5 g/dL**, >13 harmful), secondary/tertiary HPT, acidosis (bicarbonate ≥**22**), hyperkalaemia (binders, keep RAAS blockade), hypertension (chlorthalidone), ED

Not in the paper answer:

- Tertiary hyperparathyroidism (HPT) is treated with a calcimimetic, then subtotal parathyroidectomy.

Slowing progression

- Target glycated haemoglobin <7.5% and blood pressure <**130/80**; the agent matters only in proteinuric disease (RAAS blockade).
- **Sodium–glucose cotransporter-2 (SGLT2) inhibitors** protect kidney and heart and cut mortality, with or without diabetes. Add **finerenone** in CKD with type 2 diabetes.

6 · Haematuria and proteinuria: when to involve a nephrologist

- †Microhaematuria is ≥**3** red blood cells per high-power field (RBC/HPF) on microscopy; a positive dipstick needs confirmation.
- **Glomerular clues** are dysmorphic red cells, red cell casts, raised ACR or protein-to-creatinine ratio (PCR), rising SCr, hypertension and systemic disease.

ALGORITHM

1. Haematuria with or without proteinuria: send urine microscopy, ACR, SCr/eGFR.
2. If there are casts, dysmorphic cells, AKI or systemic features, refer for **urgent nephrology**.
3. If isolated non-glomerular haematuria, do a urological evaluation.
4. If isolated albuminuria, stage CKD and refer per criteria.

- †KDIGO referral criteria: AKI or an abrupt GFR fall; GFR <30; ACR ≥300 mg/g (≥30 mg/mmol); a fall >5 mL/min/1.73 m²/year; red cell casts or haematuria >20 RBC/HPF; BP refractory to ≥4 drugs; persistent potassium abnormality; recurrent stones; hereditary disease.

7 · Electrolyte and acid–base disorders

Water and sodium

- **Sodium disorders are water disorders**. **Volume depletion overrides hypo-osmolality** in driving antidiuretic hormone (ADH).
- Of 1 L, D5 leaves **83 mL** in the circulation, 0.9% saline 250 mL and 5% albumin 1000 mL.

Hyponatraemia

- †Transurethral resection (TUR) syndrome is absorption of hypotonic glycine or sorbitol during monopolar resection.

Volume	Urine	Causes
Hypovolaemic	Na >20	Diuretics, mineralocorticoid deficiency, cerebral salt wasting
Hypovolaemic	Na <20	Vomiting, diarrhoea
Euvolaemic	Osm > serum	Syndrome of inappropriate ADH (SIADH), cortisol deficiency, hypothyroidism, pain, drugs
Euvolaemic	Osm low	Polydipsia, low solute ("tea and toast", beer potomania)
Hypervolaemic	Na >20	Kidney failure
Hypervolaemic	Na <20	Heart failure, cirrhosis, nephrotic

- Treat SIADH with water restriction, then vaptans, salt with furosemide, or urea.
- **Rate**: in chronic hyponatraemia, raise sodium ≤ **8–12 mEq/L per 24 h**, or **osmotic demyelination** follows. Treat overcorrection with DDAVP and free water.
- With severe symptoms, give **3% saline to raise sodium 4–6 mEq/L**, then slow down.

Hypernatraemia

- Nephrogenic diabetes insipidus (DI) follows lithium, hypercalcaemia, hypokalaemia and **relief of obstruction**.
- Give free water. In chronic cases, lower sodium ≤ **0.5 mEq/L/h**, or cerebral oedema follows.

Potassium

P1 Management of acute hyperkalaemia — 98% intracellular; causes incl. obstructive uropathy, jejunal conduit, pseudohyperkalaemia; ECG sequence; > **6.0** urgent; binders (SZC ≈1 h, patiromer ≈7 h)

ALGORITHM

1. **Stabilise** if there is any ECG change: calcium gluconate 10%, 10 mL IV.
2. **Shift**: insulin 10 U with 50 mL of 50% dextrose; nebulised salbutamol 20 mg.
3. **Remove**: haemodialysis if oliguric or ESKD; loop diuretic if non-oliguric; bicarbonate if acidotic; a binder.
4. Stop culprit drugs.

- **Sodium polystyrene sulfonate (SPS)** is slow and causes **colonic necrosis**, so it is not first-line.
- **Hypokalaemia** from **vomiting** is caused by kidney loss, not gastric loss. Replace potassium chloride, **oral preferred**, 20–40 mEq repeated.

Calcium

- Saline and loop diuretics raise urine calcium; **thiazides lower it**.
- **Hypercalcaemia in a stone former**: non-suppressed PTH means primary HPT; if suppressed, check PTH-related peptide (renal, bladder, squamous cancer) and 1,25-vitamin D (sarcoid, lymphoma). Low urine calcium suggests familial hypocalciuric hypercalcaemia.
- Treat if symptomatic or ≥ **14 mg/dL**: **isotonic saline** 200 mL/h to urine 100–150 mL/h; loop diuretic only if overloaded; calcitonin; IV bisphosphonate; denosumab.

Acid–base

- Anion gap (AG) = Na − (Cl + HCO_3); it is raised if >12. If the AG rises less than the bicarbonate falls, a non-AG acidosis coexists.
- **Urine AG** = (Na + K) − Cl. Negative means ammonium excretion, so gut loss; positive means a renal cause (renal tubular acidosis, RTA).

P2 Renal tubular acidosis — type 1/2/4 table: defect, K, urine pH, stones; obstruction causes type 4; UAG

Not in the paper answer:

- **Topiramate and zonisamide** cause calcium phosphate stones. MCQ Laxative abuse leads to **ammonium acid urate stones**.
- †Bowel conduits and neobladders absorb ammonium chloride, causing hyperchloraemic non-AG acidosis.

8 · Urinary tract obstruction and post-obstructive diuresis

Relief of obstruction can cause dangerous fluid loss.

Term	Meaning
Hydronephrosis	Pelvicalyceal dilatation, with or without obstruction
Obstructive uropathy (OU)	Obstruction at any level
Obstructive nephropathy	Irreversible parenchymal loss

- OU is **curable**, so exclude it (†by ultrasound) in any unexplained AKI or CKD.

P1 Post-obstructive diuresis — after bilateral obstruction relief; up to **300 mL/h**, ≈2 days; 3 mechanisms (AQP2/Na-K-ATPase loss, volume expansion/ANP, urea); gradual decompression no role

ALGORITHM

1. Relieve the obstruction, then allow free oral fluids; do not give routine IV fluid.
2. If output stays high, or there is hypovolaemia, or the patient is obtunded or nil by mouth, then check a panel every **4–6 h** (Na, K, phosphate, Ca, Mg).
3. Replace ≈**75%** of the output with **0.45% saline** plus electrolytes.

KEY POINT

Why not replace all output? The diuresis is mostly appropriate excretion of retained salt, water and urea; †full replacement sustains it.

Recovery after obstruction

Canine study	Model	Recovery
Vaughan and Gillenwater	Complete unilateral ureteral obstruction	100% at 7 days; 70% at 14 days; 30% at 4 weeks; **none at 6 weeks**

- Duration, not dilatation, decides the injury. A small stone with low-grade obstruction can be observed safely for **2–4 weeks**; beyond 4 weeks the outcome is unpredictable.
- Keep a **low threshold for a stent or nephrostomy**. †Infection with obstruction needs urgent drainage.

9 · Landmark trials

Trial	Design	Result	Take-home
†PRESERVE	Bicarbonate vs saline; N-acetylcysteine (NAC) vs placebo	No difference	Saline; no NAC
†STARRT-AKI	Accelerated vs standard renal replacement therapy	No survival benefit; more dialysis dependence	Dialyse for AEIOU, not early
†DAPA-CKD / EMPA-KIDNEY	SGLT2 inhibitor, CKD with or without diabetes	Less progression	Standard in CKD
†FIDELIO-DKD	Finerenone, CKD with type 2 diabetes	Fewer kidney and cardiovascular events	Add to RAAS blockade
†CHOIR / TREAT	High vs lower haemoglobin target	Harm with high target	Avoid Hb >13 g/dL

91 Renal Transplantation

1 · End-stage disease and why transplant

Transplant beats dialysis, so plan it early.

- KDIGO stage **G5 (eGFR < 15 mL/min/1.73 m²)** is end-stage renal disease (ESRD).
- **Transplant is optimal therapy**, and **pre-emptive** transplant (before dialysis) is best. It may triple life expectancy, but death risk is **higher for the first 3–6 months**, then much lower (Wolfe).

2 · Recipient evaluation

> **KEY POINT**
> Two questions: will the patient survive the operation, and will immunosuppression unmask something dormant (infection, cancer)?

- Deceased-donor listing is allowed at eGFR ≤ **20**.

Recurrence risk	Diseases	Implication
Very low	Autosomal dominant polycystic kidney disease (ADPKD), dysplasia, Alport, **genetic** focal segmental glomerulosclerosis (FSGS)	None
Higher	**Primary FSGS**, membranoproliferative GN, membranous	Counsel; monitor
Histological often, clinical rarely	IgA nephropathy	—
Metabolic	Primary hyperoxaluria	Combined liver–kidney

Disease coded in native-kidney genes cannot recur in the graft; disease from circulating or immune factors can.

- **Cardiac**: coronary artery disease is the main cause of death; screen high-risk patients.
- **Vascular**: heavy iliac calcification may prevent anastomosis (non-contrast CT). ADPKD with a family history of intracranial bleed needs an aneurysm screen.
- **Obesity** is a relative contraindication that increases wound complications, delayed graft function (DGF) and rejection.
- **Infection**: active tuberculosis contraindicates until treated. Screen for hepatitis B, C and HIV, and **vaccinate before** transplant, because live vaccines are barred afterwards.

Malignancy	Transplant
Prostate Gleason ≤6; superficial non-melanoma skin cancer; incidental renal tumour <1 cm	Without prior treatment
Basal/squamous skin, melanoma in situ, renal cell carcinoma <3 cm, small low-grade thyroid, superficial bladder cancer	No wait after cure
All other	After treatment, individualised wait

P2 Pre transplant nephrectomy — indications: ADPKD space/infection/bleeding/pain, chronic infection, VUR with UTI, heavy proteinuria

3 · Immunological compatibility

P3 Discuss the role of tissue typing and cross match in live and cadaveric — ABO rules (A2 donors), HLA class I/II, sensitisation, crossmatch, Luminex virtual crossmatch, paired exchange

4 · Deceased donation

P2 P1 Deceased (cadaver) kidney donor: selection, identification, maintenance, distribution and retrieval; Kidney retrieval techniques for organ donation — DBD vs DCD, contraindications, KDPI/EPTS, en bloc < **15 kg**, retrieval steps, UW/HTK, machine perfusion ↓DGF

Not in the paper answer:

- En bloc kidneys go to an adult with body mass index <30 and low immunological risk; thrombosis is 10–15%.

	Donation after brain death (DBD)	Donation after circulatory death (DCD)
Warm ischaemia	Minimal	Present, so rapid aortic cannulation and cooling are needed
Preservation	Static cold storage	Usually machine perfusion (reduces DGF)

ALGORITHM

1. Midline incision and sternotomy; mobilise bowel off the retroperitoneum.
2. Control the aorta and inferior vena cava (IVC).
3. Find aberrant or accessory arteries **before cannulation**.
4. Perfuse; ice via opened Gerota fascia.
5. Remove kidneys en bloc with **periureteric tissue preserved**.
6. Separate on the back table; store cold.

5 · Living donation

P2 Pre-operative evaluation of the living (related and unrelated) renal donor — donor safety, exclusions (bilateral stones), GFR by 2 methods, CT angiography, split function; function 60–80%, ESKD 90 vs 14/10,000

Not in the paper answer:

- Donor mortality is **3/10,000**. An independent donor advocate is mandatory.
- **Laparoscopic donor nephrectomy** is standard.

6 · The operation

- On the back table, ligate hilar lymphatics (lymphocele). **Never dissect the triangle between lower pole and ureter.** Extend a short right vein with donor IVC.
- **Multiple arteries** (18–43%) mean more DGF but similar survival; use a Carrel patch, conjoining or separate anastomoses.
- **Always keep the lower pole artery**, because it supplies the ureter; upper pole branches ≤2 mm may be ligated.
- The graft goes in the retroperitoneal iliac fossa. **Contralateral placement** brings the pelvis and ureter anterior.

	Gibson	Hockey stick
Course	Oblique, 2 cm above pubis to near the anterior superior iliac spine (ASIS)	Midline suprapubic, up the rectus edge
Access	Iliac vessels	Better to common iliac, IVC
Hernia	Lower	Higher; rectus denervation

ALGORITHM

1. **Vein first**: end-to-side to the external iliac vein, 5-0/6-0 polypropylene.
2. Artery end-to-side to the external iliac artery (punch arteriotomy; avoid calcified clamp sites).
3. Reperfuse, then position the kidney without kink or stretch.

- A flaccid graft suggests poor inflow and a tense, engorged graft poor outflow; check with intraoperative ultrasound.

ALGORITHM

1. **Lich–Gregoir** extravesical: fill the bladder; incise the detrusor down to mucosa.
2. Open the mucosa distally and spatulate the ureter.
3. Suture mucosa to mucosa with 5-0/6-0 absorbable, with or without a stent.
4. Close the detrusor loosely as a short tunnel.

- Document stent removal, because forgotten stents encrust.

7 · Surgical complications

Early vascular problems lose grafts within hours.

P1 Vascular complications after renal transplantation, including transplant renal artery stenosis — arterial/venous thrombosis (reversed diastolic flow), TRAS (parvus-tardus, angioplasty first), thrombosis risks, torsion, pseudoaneurysm, biopsy AV fistula

Not in the paper answer:

- Bleeding (brisk drain, falling haematocrit) needs transfusion and re-exploration.

P1 Lymphocele and postoperative lymphatic leak (chylous ascites) — risks incl. mTOR inhibitors; lymph creatinine = serum; treat if symptomatic; laparoscopic fenestration best, aspiration worst

Transplant nephrectomy

- Indications are **graft intolerance syndrome** (pain, swelling, haematuria, fever; steroids first), infection and tumour, not an asymptomatic failed graft.
- **Early (<6–8 weeks), go extracapsular**; later, go **subcapsular**, except for tumour.

8 · Rejection

Chronic antibody injury now limits graft survival.

- The usual regimen is induction, then tacrolimus and mycophenolate, with or without steroids.

P1 Renal allograft rejection. — hyperacute/acute/subclinical/chronic; acute <10%, chronic AMR limits survival; biopsy (Banff TCMR/AMR); risks; de novo DSA, dd-cfDNA NPV >90%; TCMR steroids, then ATG; AMR plasmapheresis, IVIG, rituximab, eculizumab, anti-IL-6

Not in the paper answer:

- De novo donor-specific antibody (DSA) is linked to class II mismatch and under-immunosuppression.

- Hyperacute rejection is †irreversible, so the graft is removed.

9 · Landmark trials

Trial	Design	Result	Take-home
Wolfe 1999	Registry: transplant vs wait-listed dialysis	Risk 2.84 early; equal at 106 days; long-term 0.32	Early peril, durable survival gain

★ Putting it together: kidney function and transplant decisions

A · Rising creatinine after urological surgery

ALGORITHM

1. **Postrenal first:** check the catheter and scan. After cystectomy, a rise with ascites may be a urine leak (drain creatinine ≫ serum).
2. **Prerenal vs intrinsic:** FeNa <1% with bland urine means volume depletion; >1% with granular casts means acute tubular necrosis. Stop nephrotoxins, RAAS blockers and metformin.
3. **Dialysis** only for refractory "AEIOU": acidosis, electrolytes (K), intoxication, overload, uraemia.

B · Post-obstructive diuresis

KEY POINT

After bilateral or solitary-kidney relief, monitor output and electrolytes. Diuresis is mostly physiological; †it is pathological if it persists after euvolaemia. †Do not replace every millilitre; if the patient cannot drink, replace ≈75% with 0.45% saline.

C · Transplant ureteric complications

Problem	Clue	Management
Urine leak (early)	High drain output; fluid creatinine ≫ serum	Foley and nephrostomy ± stent; explore if large or persistent
Ureteric stricture	Falling function ± mild hydronephrosis	Nephrostomy, then dilation or endoureterotomy; open repair is most durable
Lymphocele	Fluid creatinine ≈ serum; may obstruct ureter or vessels	Treat only if symptomatic (<15%); laparoscopic window is best
Stent infection	Colonised stent	Remove at 14–21 days (earlier risks leak; later risks infection)

Prevent them with a stent and by sparing the lower pole–ureter triangle.

D · Rejection at a glance

Type	Cause	Point
Hyperacute	Preformed donor-specific antibody (DSA)	Crossmatch prevents it; †irreversible, so remove graft
Acute	T-cell-mediated (TCMR) or antibody-mediated (AMR), Banff	Biopsy. TCMR: IV steroids, then antithymocyte globulin (ATG). AMR: steroids, IV immunoglobulin, plasmapheresis, rituximab
Chronic	Mainly antibody-mediated	Main late loss; subclinical

E · Child going for transplant

Fix the bladder before transplant.

ALGORITHM

1. Most have congenital anomalies (CAKUT), chiefly valves.
2. Make the bladder safe: age-appropriate capacity at detrusor pressure <40 cm H_2O (antimuscarinics, catheterisation, augmentation).
3. Remove refluxing ureters or kidneys causing infection, polyuria or hypertension.
4. A poor bladder destroys the graft.